DO NOW — RECALL

Programming Techniques · Sequence and Selection

Name: _____ Class: _____ Date: _____

1. What does a full adder include that a half adder does not?

2. What is a local variable?

3. Why does a stack use LIFO?

4. Simplify A ∨ (A ∧ B).

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Carry-in

2. Model answer

A variable declared within a scope

3. Model answer

The most recently inserted item is removed first

4. Model answer

A, by absorption

By the end, you can…

• Trace instructions in sequence

• Choose IF/ELSE conditions

• Write a short C# decision algorithm

Vocabulary — know these terms

Recall the term — sequence

Recall the term — selection

Recall the term — Boolean

Recall the term — condition

Recall the term — trace

NEW LEARNING

Read, annotate and explain

Follow the order of execution

int score = 7; score = score + 3; if (score >= 10) Console.WriteLine("Pass");

Explain the important decision

Sequence changes score before selection tests it. The condition is true at exactly 10 because >= includes the boundary.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Follow the order of execution

Update in sequence

score starts at 7. Adding 3 changes the same variable to 10; the condition has not been tested yet.

Evaluate the condition

Substitute the new value: 10 >= 10 is true. Using the old value 7 would trace the wrong state.

Take the selected branch

The true branch prints Pass. For initial scores 6, 7 and 8, the updated scores are 9, 10 and 11: only the latter two print Pass.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Problem 1

value = 3; value *= 4; is value > 10?

Method / working:

2. Problem 2

A user enters age 16. Test age >= 18.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Trace x = 4; x = x + 7; output x. [2]

Question 2

Apply independently
Write a C# condition that prints Adult when age >= 18. [3]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Explain one difference between sequence and selection. [2]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: an else branch runs when its if condition is false.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.